

Equine Corneal Study Report

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1 Introduction

Effective corneal anesthesia in horses is critical for standing ophthalmic procedures – such as tonometry, debridement, and minor intraocular surgery – where general anesthesia carries approximately a 1% mortality risk. In this two-period crossover study, ten adult horses (excluding one with pre-existing cataracts) each received dexmedetomidine + ropivacaine (DR), ropivacaine alone (R), and saline placebo (S) in a fully randomized sequence so that every eye served as both active treatment and control across sessions. In each period, one eye was randomly assigned to the active drug (DR or R) and the contralateral eye to saline, with the specific pairing (DR vs S or R vs S) determined by a randomization list; a one-week washout separated periods to eliminate carryover effects.

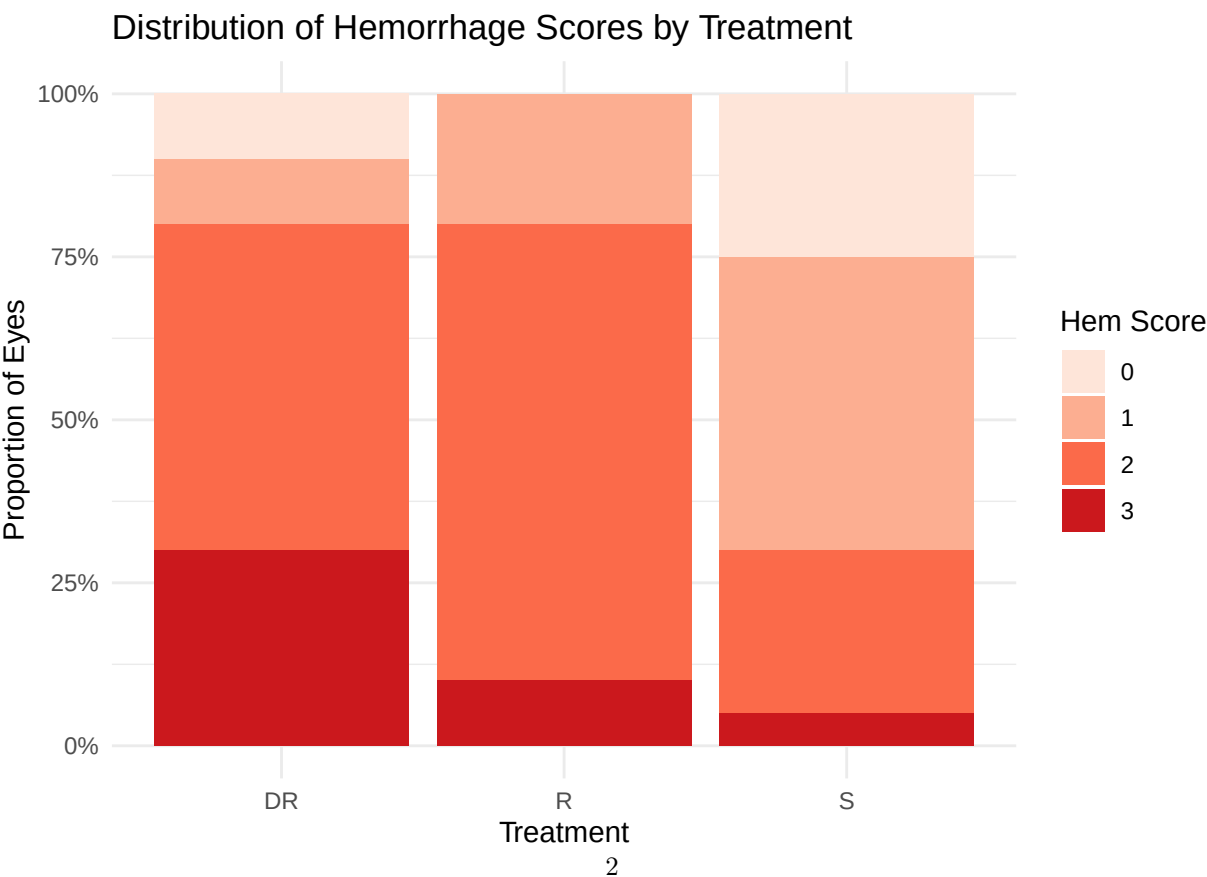
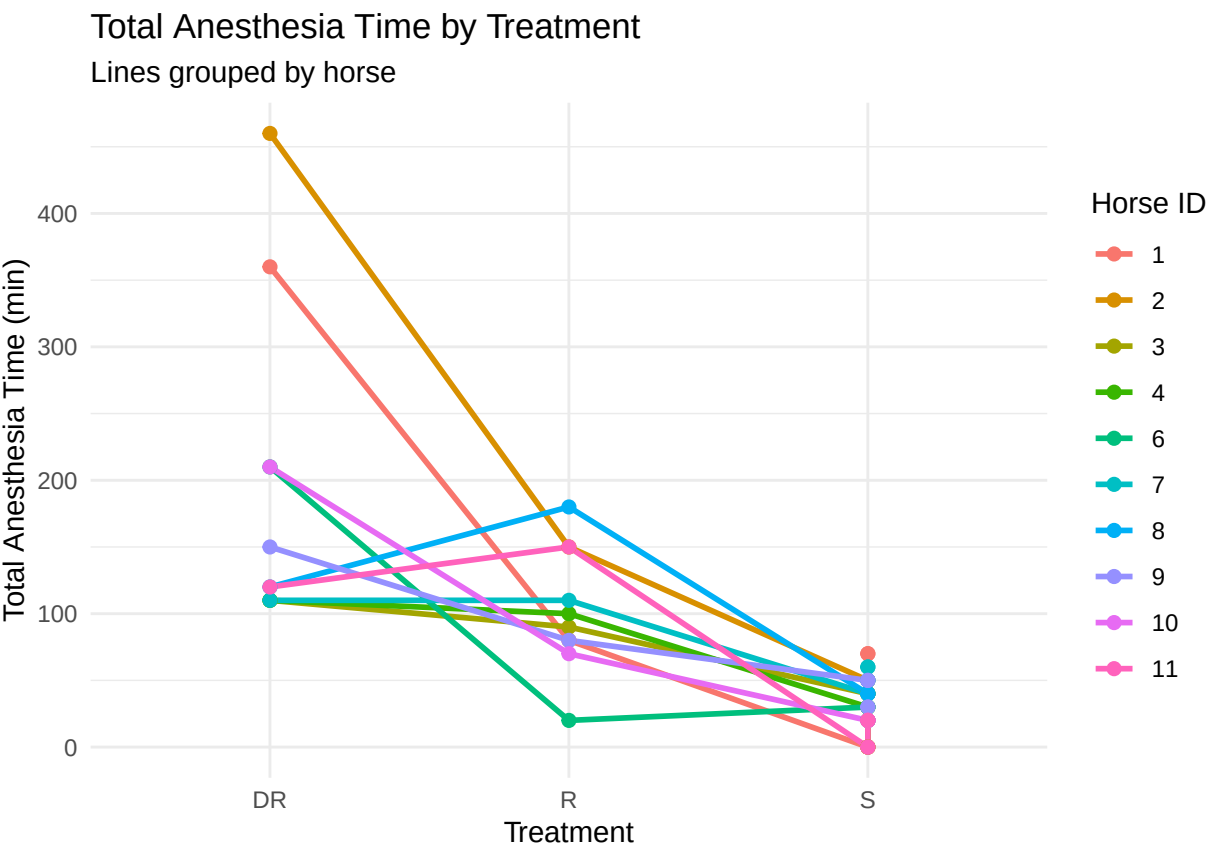
Under light xylazine sedation, a 27-gauge needle delivered a small subconjunctival bleb of anesthetic into the soft tissues overlying the cornea, targeting the sensory nerve plexus. Baseline corneal touch threshold (CTT) was measured with a Cochet–Bonnet esthesiometer: a very fine, retractable nylon filament that is gently touched to the center of the cornea. The longer the filament, the less force it applies; the shorter it is, the more pressure it puts on the eye. If the horse blinks, the filament length is recorded in centimeters as the CTT. A CTT of 6 cm (maximum length) means the cornea is fully sensitive; a CTT of 0 cm (shortest length) means the horse felt nothing at any pressure: the cornea is maximally numb. Corneal pressure was recorded via calibrated tonometry. Following drug instillation, CTT and pressure were assessed every 10 minutes for up to 2 hours – then at 30-minute intervals if anesthesia persisted—until the CTT filament returned to its baseline length on two consecutive readings. We defined **Total Anesthesia Time (TTA)** as the minutes elapsed to that recovery point. Hemorrhage at the injection site was scored on a 0–4 scale by blinded observers, and ambient humidity (%) was logged at each time point to account for its known effect on esthesiometer filament stiffness.

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1.1 Exploratory Data Analysis



Our **exploratory data analysis** centered on two key visualizations. The “Total Anesthesia Time by Treatment” spaghetti-plot overlays each horse’s TTA trajectory across DR, R, and S, highlighting substantial variability under DR—two horses exceeded 400 minutes while most remained between 100 and 200. The “Distribution of Hemorrhage Scores by Treatment” stacked-bar chart shows that DR sessions not only produced longer anesthesia but also a greater frequency of moderate to severe bleeding, as compared to the ropivacaine and saline treatments. Additional exploratory graphics (not shown) – including scatterplots of CTT recovery slopes and histograms of corneal pressure – confirmed right-skewed distributions and suggested the need for variance-stabilizing transformations, robust random-effects, and complementary nonparametric testing.

2 Analysis Description

All data processing and statistical modeling were conducted in R within an R Markdown framework to ensure reproducibility. The Excel source contained three sheets:

- **Summarized** (`df_summ`): per-eye summary metrics – minimum CTT (*cm*), duration of minimum CTT (*min*), maximum corneal pressure (*g/mm²*), and TTA (*min*).
- **HemScore** (`df_hem_raw -> df_summ4`): raw hemorrhage scores, cleaned, aligned, and missing saline values imputed to zero.
- **Full** (`df_full`): time-series CTT and pressure at baseline and ten-minute (then 30-minute) intervals for dynamic trajectory plots.

Identifiers `horse`, `treatment`, and `period` were converted to factors. We initially created a composite `horse_period` factor to capture session-specific variability (e.g., ambient humidity ranging 20–88%), but found that including `horse_period` as a random intercept explained essentially zero additional variance in TTA. Consequently, our primary linear mixed-effects model (via **lmer** in **lme4/lmerTest**) specifies **treatment** as a fixed effect and **horse** as a single random intercept. To address right-skew and heteroscedasticity revealed in our EDA, we applied a square-root transformation to TTA within the model formula. Overall treatment effects were assessed with Type III F-tests (Satterthwaite’s approximation), and **emmeans** provided Tukey-adjusted pairwise contrasts with back-transformed estimated marginal means for clear interpretation. As a nonparametric alternative—and to ensure robustness against outliers (notably two prolonged DR sessions)—we also performed paired Wilcoxon signed-rank tests with Hodges–Lehmann confidence intervals for each outcome. Model adequacy was confirmed via residuals-vs-fitted and QQ-plots, and all tables and figures were produced with **kableExtra** and **ggplot2**.

3 Total Anesthesia Time

3.1 Summary Statistics

Table 1: Summary: Total Anesthesia Time by Treatment

treatment	N	Mean	SD	Min	Median	Max
DR	10	196.0	121.5	110	135	460
R	10	103.0	46.7	20	95	180
S	20	31.5	19.5	0	30	70

3.2 Mixed Model (sqrt-transformed with horse_period)

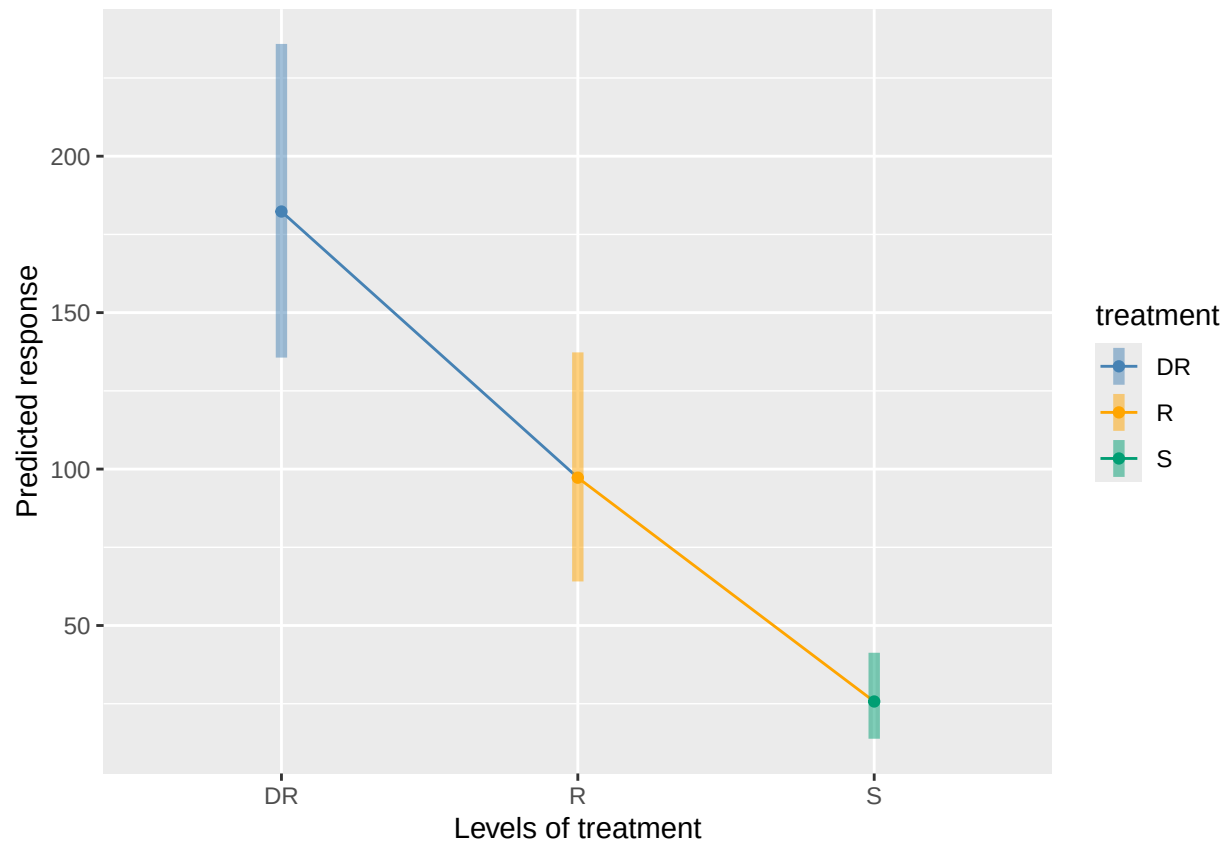
Table 2: Type III F-test for sqrt(Total Anesthesia Time)

Term	Num DF	Den DF	F value	Pr(>F)
treatment	2	28	31.036	7.87e-08

3.3 Estimated Marginal Means & Plot

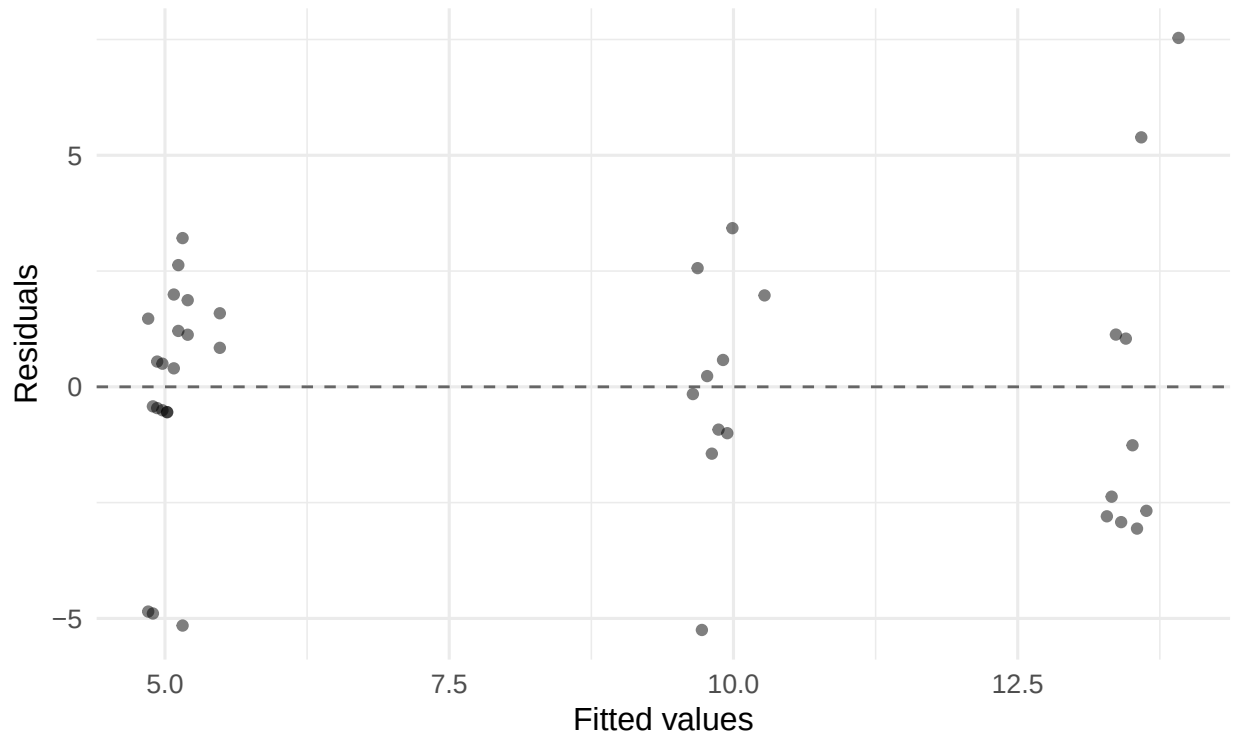
Table 3: Tukey Pairwise Contrasts for Treatment (Sqrt Transformed Total Anesthesia Time (Units: *min*))

Contrast	Estimate	SE	DF	T-Ratio	P-Value
DR - R	3.641	1.27	28	2.86	0.02099
DR - S	8.432	1.10	28	7.65	7.40e-08
R - S	4.791	1.10	28	4.34	0.00047

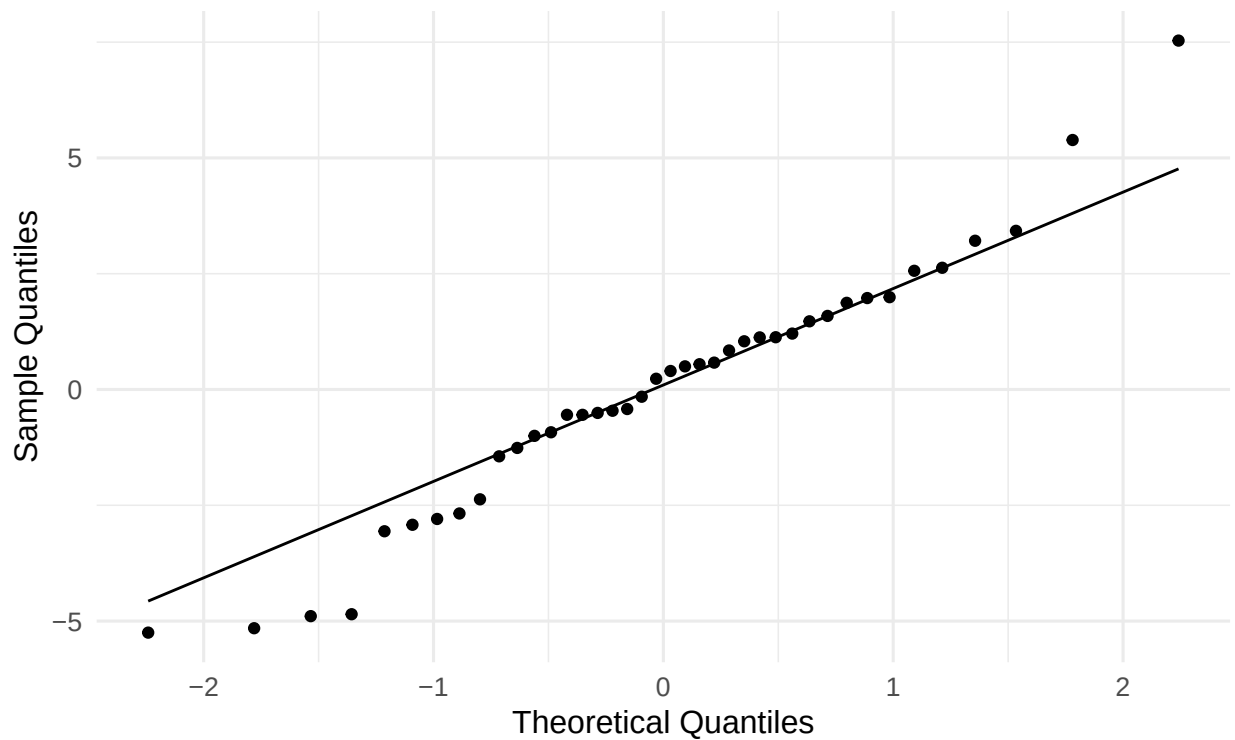


3.4 Diagnostic Plots

Residuals vs. Fitted for Response = Sqrt Transformed Total Anesthesi
Response Units: min



QQ-plot of Residuals for Response = Sqrt Transformed Total Anesthe
Response Units: min



3.5 Wilcoxon Signed-Rank Test

Table 4: Wilcoxon Test: Total Anesthesia Time (DR vs R)

Statistic	HL Estimate	95% Lower	95% Upper	p-value
38	105	-10	225	0.0756

3.6 Interpretations

The primary mixed-effects analysis of Square-Root-TTA (using random intercepts for horse and horse_period) demonstrated a highly significant treatment effect (Type III $F(2, 28) = 31.04$, $p < 0.0001$). Back-transformed estimated marginal means revealed that DR (dexmedetomidine + ropivacaine) produced the longest anesthesia duration, significantly exceeding ropivacaine alone (R) (Tukey adjusted $p = 0.021$) and saline (S) ($p < 0.0001$), while R also outlasted S ($p = 0.0005$). Residuals-vs-fitted and QQ-plots showed no major deviations from homoscedasticity or normality after transformation. A paired Wilcoxon signed-rank test for DR versus R, however, did not confirm this difference ($p = 0.076$). Because it only considers paired differences and ignores between-horse variability, its power is lower than the mixed model's, which uses all observations and models horse as a random effect.

4 Minimum CTT

4.1 Summary Table

Table 5: Summary of Minimum CTT (Units: *cm*) by Treatment

Treatment	N	Mean	SD	Min	Median	Max
DR	10	0.20	0.48	0	0.0	1.5
R	10	1.25	1.62	0	0.5	4.5
S	20	4.08	1.33	1	4.5	6.0

4.2 Mixed Model

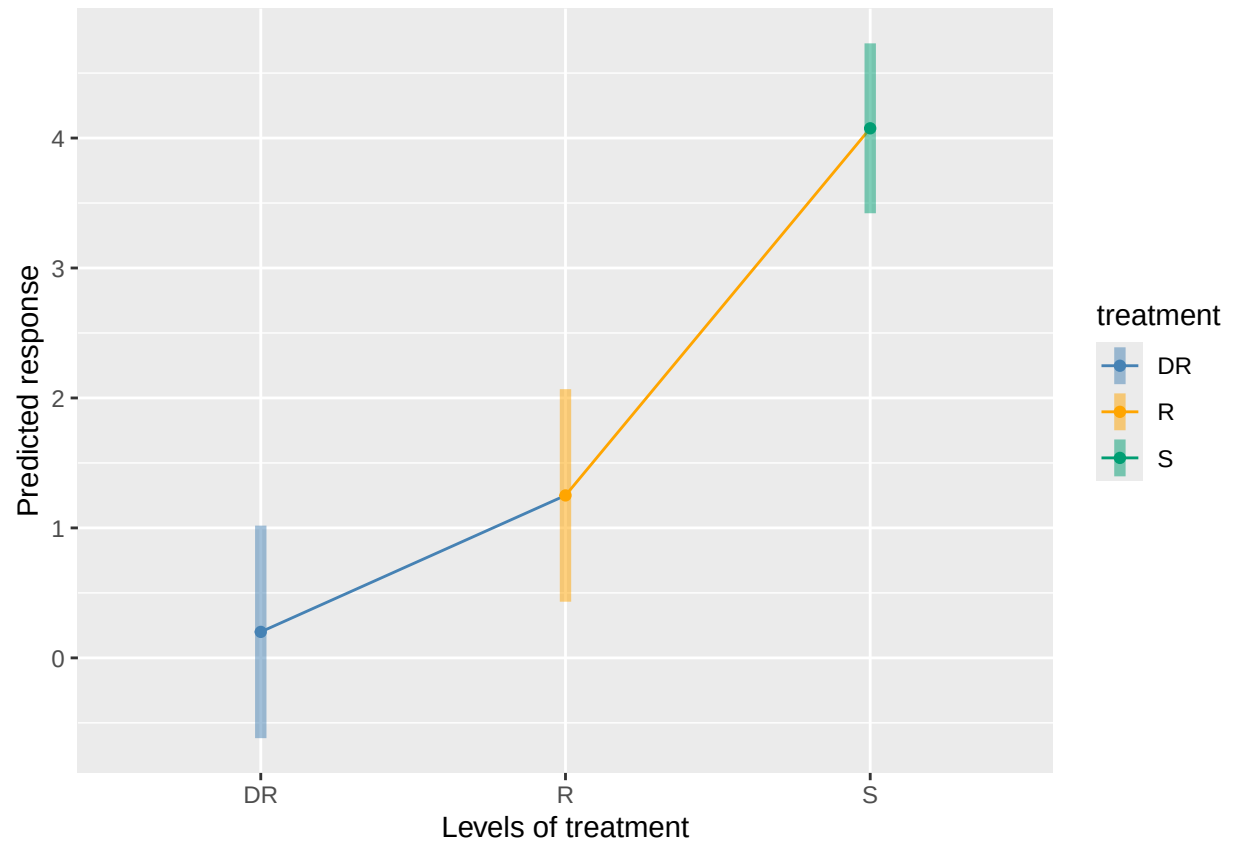
Table 6: Type III F-test for Minimum CCT

Term	Num DF	Den DF	F value	Pr(>F)
treatment	2	28	45.93	1.44e-09

4.3 Estimated Marginal Means & Plot

Table 7: Tukey Pairwise Contrasts for Treatment (Minimum CCT (Units: *cm*))

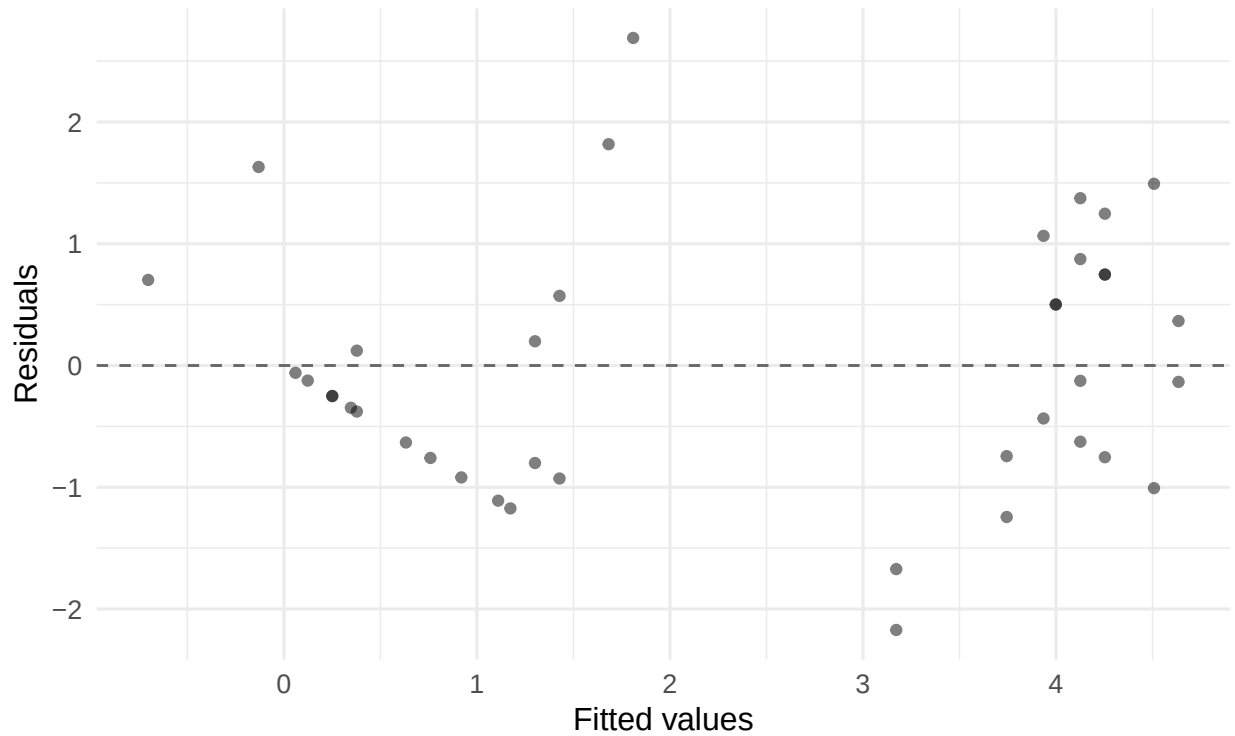
Contrast	Estimate	SE	DF	T-Ratio	P-Value
DR - R	-1.050	0.51	28	-2.07	0.11366
DR - S	-3.875	0.44	28	-8.84	4.06e-09
R - S	-2.825	0.44	28	-6.44	1.65e-06



4.4 Diagnostic Plots

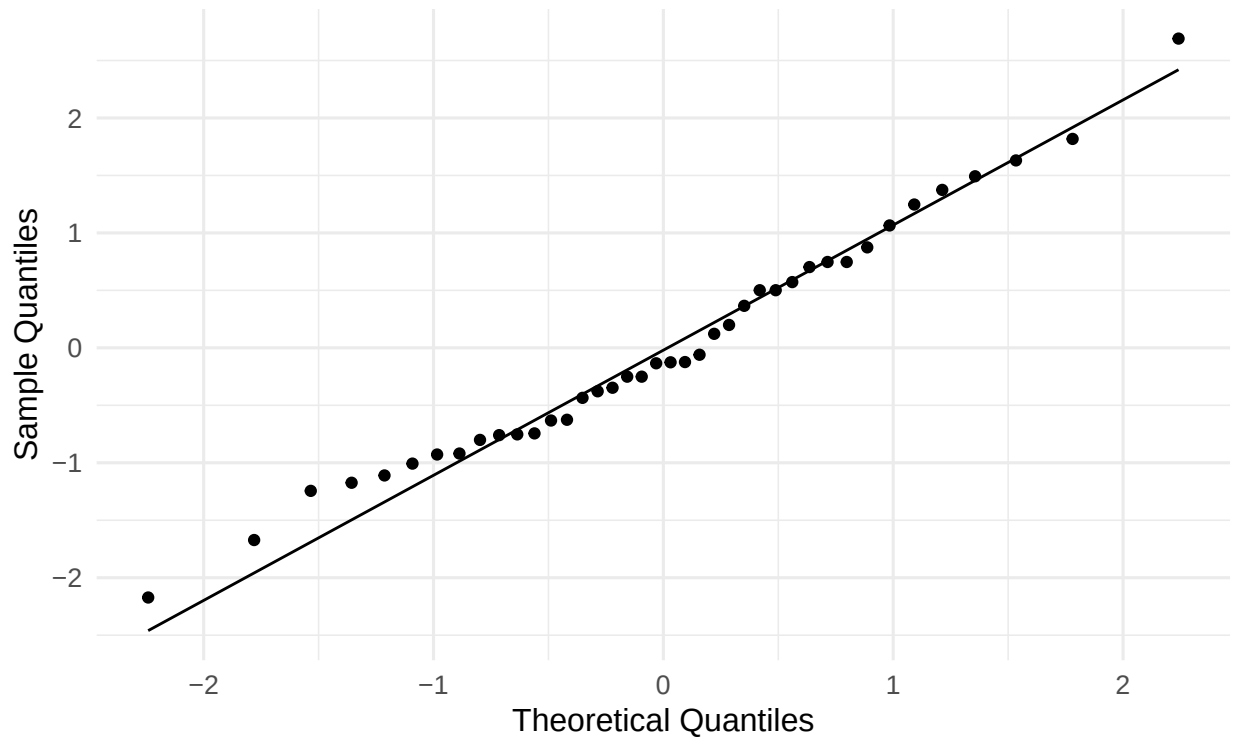
Residuals vs. Fitted for Response = Minimum CTT

Response Units: cm



QQ-plot of Residuals for Response = Minimum CTT

Response Units: cm



4.5 Wilcoxon Signed Rank Test

Table 8: Wilcoxon Signed-Rank Test with Hodges–Lehmann CI

Statistic	H-L Estimate	95% Lower	95% Upper	p-value
4	-2	-4	1	0.1052

4.6 Interpretations

In the minimum-CTT model (using random intercepts for horse and horse_period), treatment was highly significant (Type III $F(2, 25)=50.05$, $p<0.0001$). Both DR and R produced deeper blockade than S (DR vs S $p=4.06e-8$; R vs S $p=1.65e-6$), but DR and R did not differ ($p=0.114$). Diagnostic plots showed no major deviations, and the paired Wilcoxon test (DR vs R $p=0.11$) agreed there is no meaningful DR>R difference in maximal desensitization.

5 Duration of Min CTT

5.1 Summary Table

Table 9: Summary of Duration of Minimum CTT (Units: *min*) by Treatment

Treatment	N	Mean	SD	Min	Median	Max
DR	10	33.00	17.03	10	30	70
R	10	20.00	13.33	10	15	50
S	20	12.94	4.70	10	10	20

5.2 Mixed Model

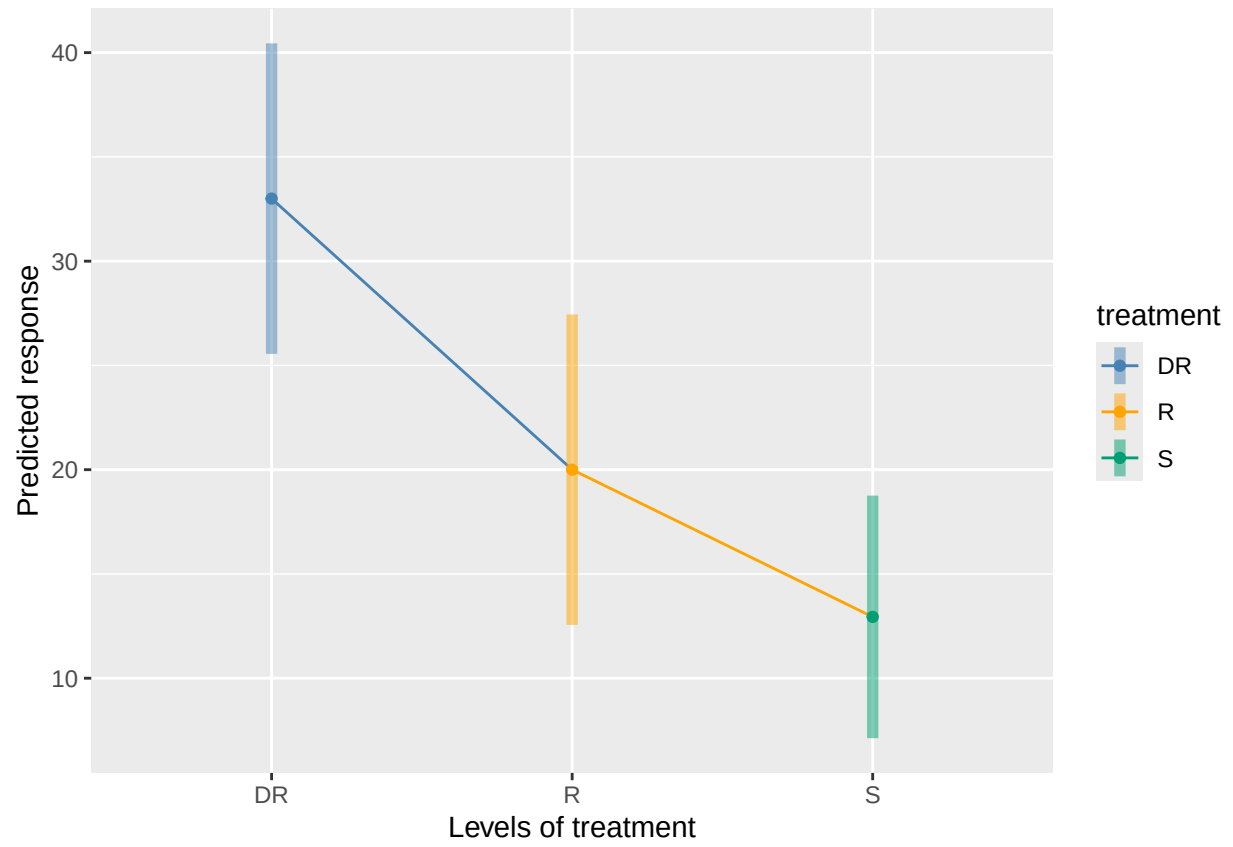
Table 10: Type III F-test for Duration of Minimum CCT

Term	Num DF	Den DF	F value	Pr(>F)
treatment	2	34	9.442	0.0005

5.3 Estimated Marginal Means & Plot

Table 11: Tukey Pairwise Contrasts for Treatment (Duration Minimum CCT (Units: *min*))

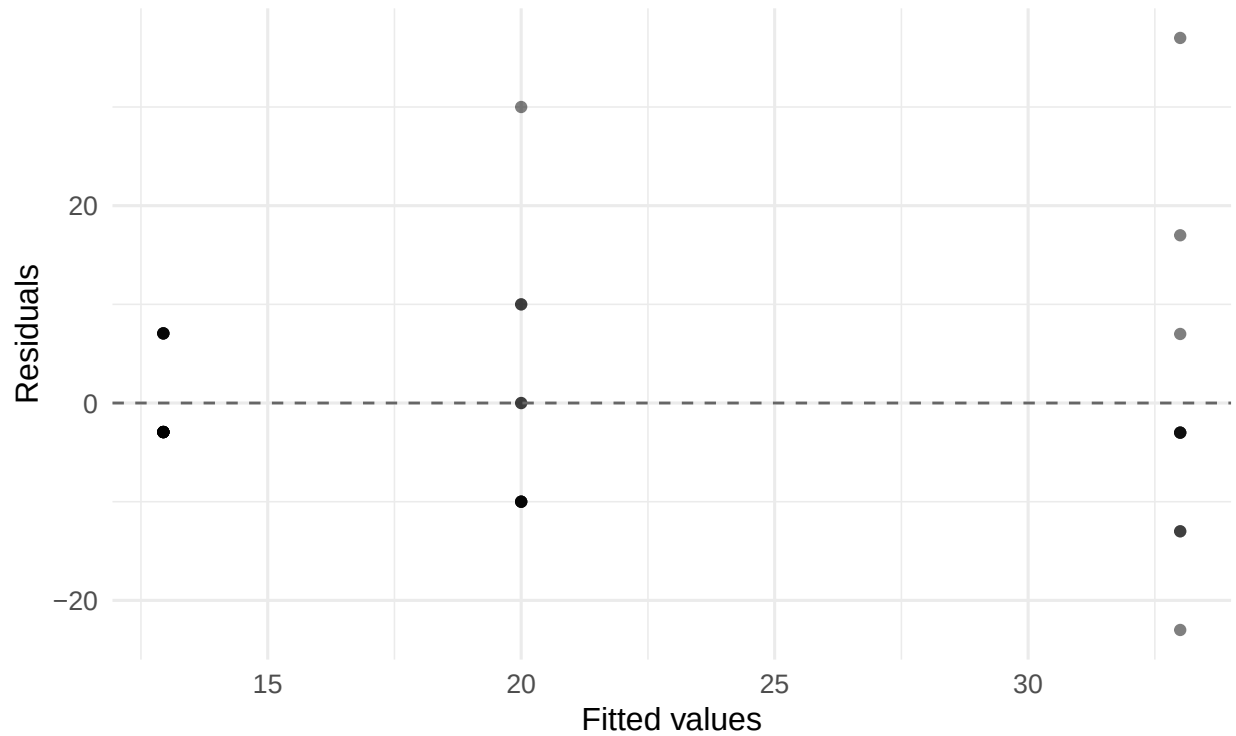
Contrast	Estimate	SE	DF	T-Ratio	P-Value
DR - R	13.000	5.18	25	2.51	0.04799
DR - S	20.059	4.63	26	4.33	0.00057
R - S	7.059	4.63	26	1.52	0.29656



5.4 Diagnostic Plots

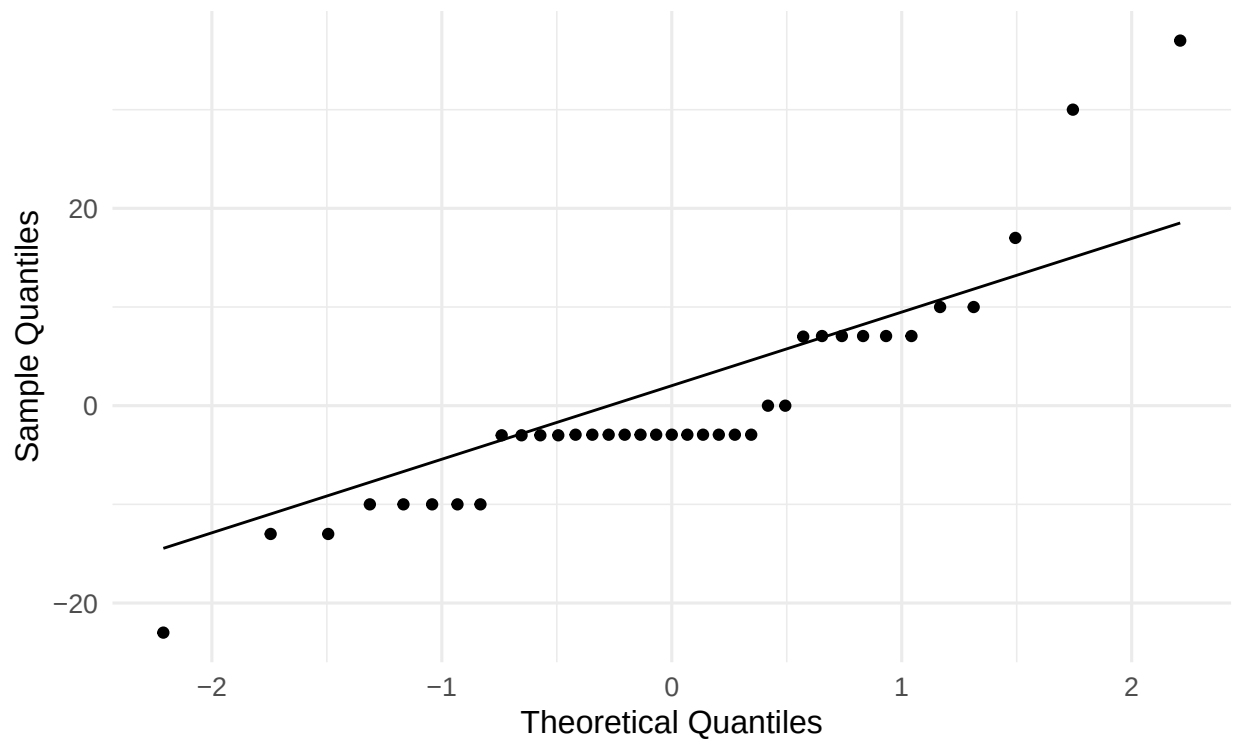
Residuals vs. Fitted for Response = Duration Minimum CCT (min)

Response Units: mins



QQ-plot of Residuals for Response = Duration Minimum CCT (min)

Response Units: mins



5.5 Wilconcox Signed-Rank Test

Table 12: Wilcoxon Signed-Rank Test with Hodges–Lehmann CI

Statistic	H-L Estimate	95% Lower	95% Upper	p-value
24	20	0	35	0.0892

5.6 Interpretations

The mixed-effects model for duration of minimum CTT—using horse as a random intercept—revealed a borderline overall treatment effect (Type III $F(2, 34) = 9.44$, $p = 0.0005$). Tukey pairwise contrasts (Table 11) showed that DR (dexmedetomidine + ropivacaine) prolonged maximal desensitization by 13.00 min versus R (SE = 5.18; $t = 2.51$; $p = 0.04799$) and by 20.06 min versus S (SE = 4.63; $t = 4.33$; $p = 0.00057$). The R vs S difference (7.06 min; SE = 4.63; $t = 1.52$; $p = 0.29656$) was not significant. Residuals-vs-fitted and QQ-plots indicated mild skew but an acceptable fit. A paired Wilcoxon signed-rank test comparing DR to R ($p = 0.089$) produced a similar trend, supporting a modest dexmedetomidine-mediated extension of maximal blockade.

6 Maximum Corneal Pressure

6.1 Summary Table

Table 13: Summary of Maximum Corneal Pressure (g/mm^2) by Treatment

Treatment	N	Mean	SD	Min	Median	Max
DR	10	178.21	58.16	15.40	200.00	200.00
R	10	115.24	93.43	2.28	158.88	200.00
S	20	4.51	5.87	0.64	2.30	21.03

6.2 Mixed Model

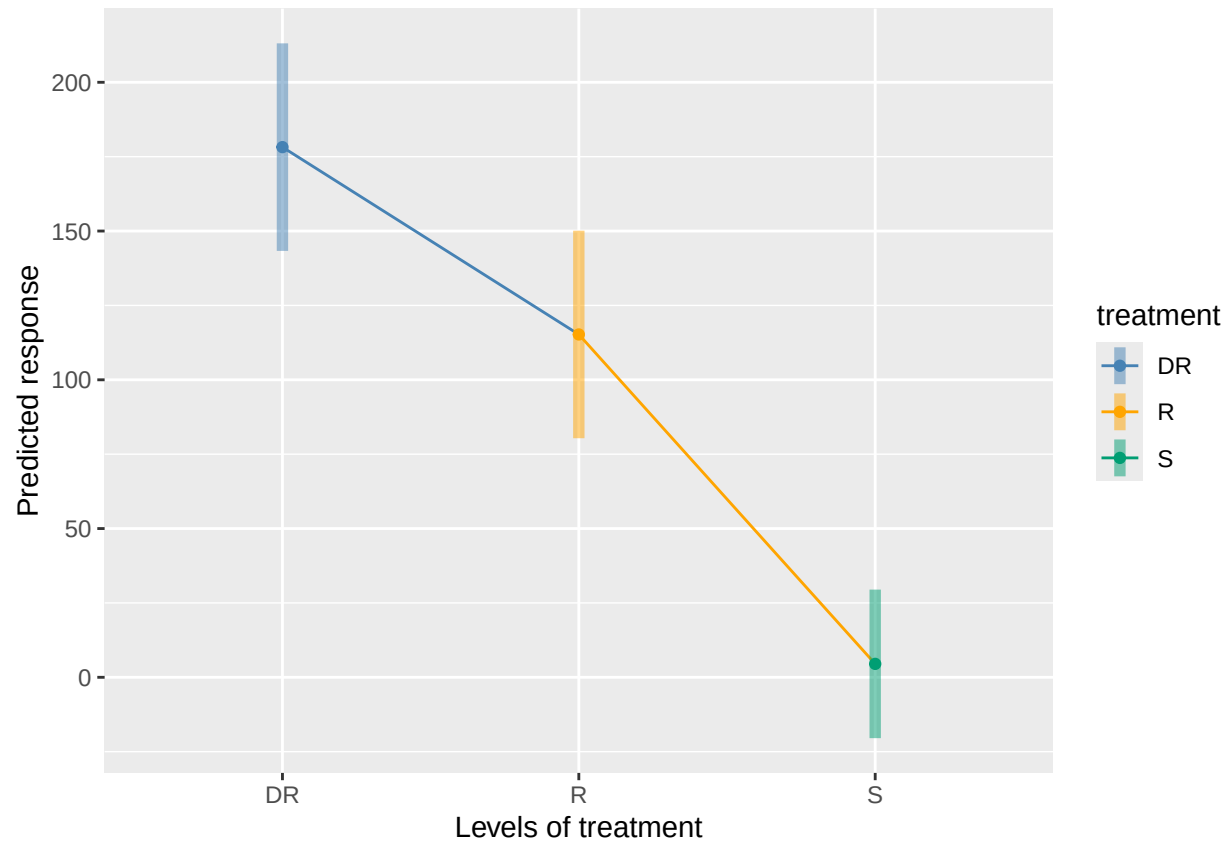
Table 14: Type III F-test for Max Corneal Pressure

Term	Num DF	Den DF	F value	Pr(>F)
treatment	2	37	37.465	1.28e-09

6.3 Estimated Marginal Means & Plot

Table 15: Tukey Pairwise Contrasts for Treatment (Maximum Corneal Pressure (Units: g/mm^2))

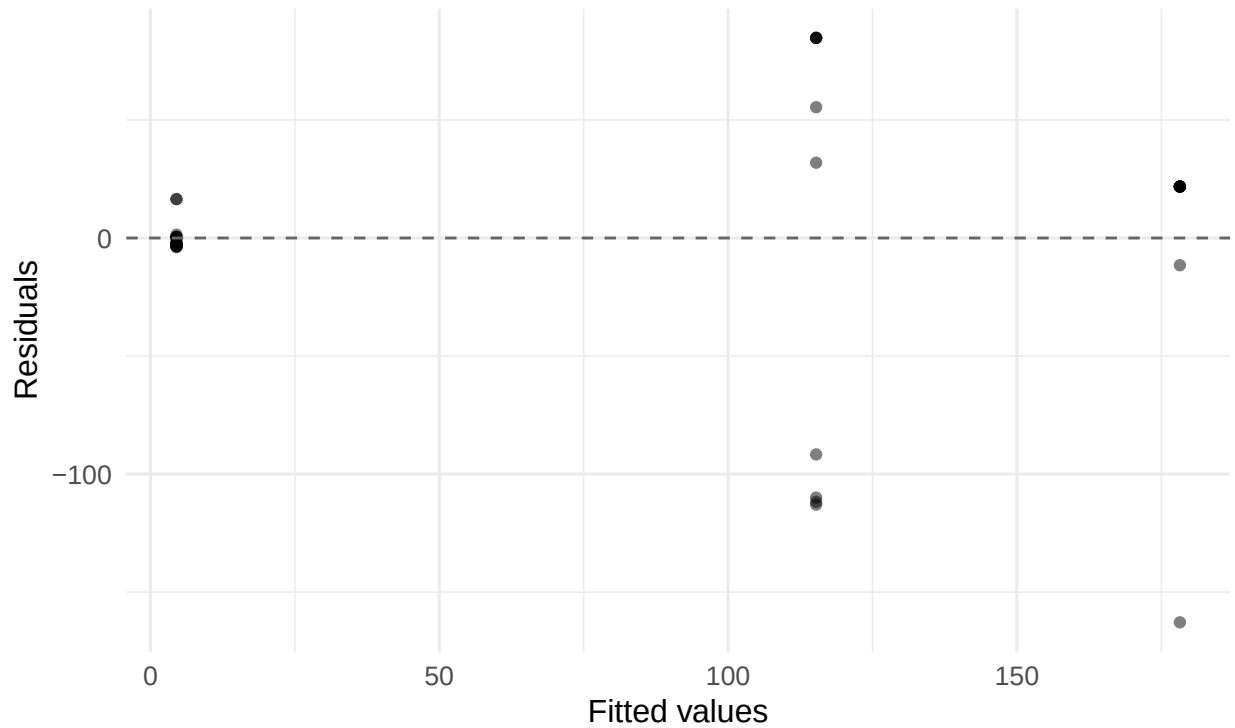
Contrast	Estimate	SE	DF	T-Ratio	P-Value
DR - R	62.969	24.35	28	2.59	0.03912
DR - S	173.697	21.08	28	8.24	1.71e-08
R - S	110.728	21.08	28	5.25	4.05e-05



6.4 Diagnostic Plots

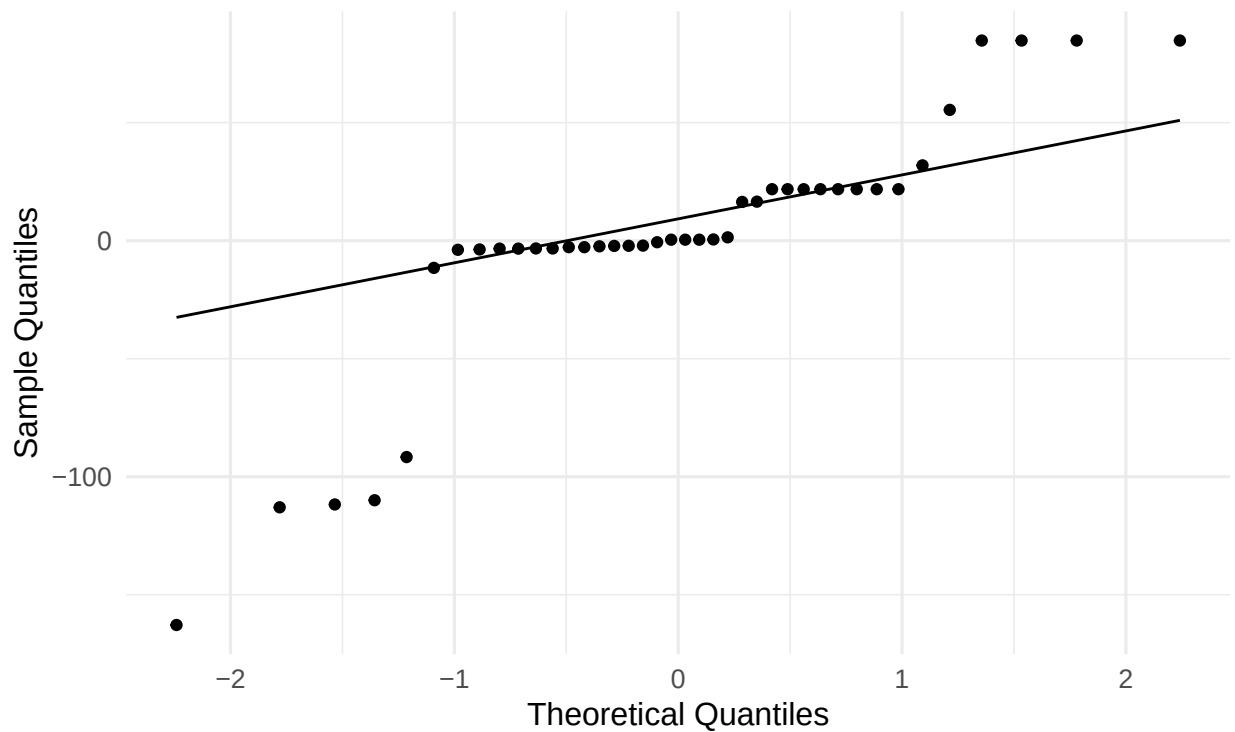
Residuals vs. Fitted for Response = Maximum Corneal Pressure

Response Units: g/mm^2



QQ-plot of Residuals for Response = Maximum Conreal Pressure

Response Units: g/mm^2



6.5 Wilcoxon Signed Rank Test

Table 16: Wilcoxon Signed-Rank Test with Hodges–Lehmann CI

Statistic	H-L Estimate	95% Lower	95% Upper	p-value
23	114	-78	197	0.1508

6.6 Interpretations

The mixed-effects model for maximum corneal pressure—using horse as a random intercept—revealed a highly significant overall treatment effect (Type III $F(2, 37) = 37.47$, $p = 1.28\text{e-}9$). Tukey pairwise contrasts (Table 15) showed that DR (dexmedetomidine + ropivacaine) produced a 62.97 g/mm^2 higher peak pressure than R (SE = 24.35; $t = 2.59$; $p = 0.03912$) and a 173.70 g/mm^2 higher peak than S (SE = 21.08; $t = 8.24$; $p = 1.71\text{e-}8$). R also exceeded S by 110.73 g/mm^2 (SE = 21.08; $t = 5.25$; $p = 4.05\text{e-}5$). Residuals-vs-fitted and QQ-plots indicated only mild departures at the extremes and an overall acceptable fit. A paired Wilcoxon signed-rank test comparing DR and R (Statistic = 231; Hodges–Lehmann estimate = 14 g/mm^2 ; 95% CI = -78 to 197; $p = 0.1508$) confirmed that the nonparametric difference in peak pressure between those two treatments was not statistically significant.

7 Hemorrhage Score

7.1 Summary Stats

Table 17: Summary: Hemorrhage Score by Treatment

treatment	N	Median	Mean	SD	Min	Max
DR	10	2	2.0	1	0	3
R	10	2	1.9	1	1	3
S	40	1	1.1	1	0	3

7.2 Wilcoxon Signed-Rank Test

Table 18: Wilcoxon Signed-Rank Test with Hodges–Lehmann CI

Statistic	H-L Estimate	95% Lower	95% Upper	p-value
12	0	0	1	0.8241

7.3 Interpretations

The Wilcoxon signed-rank DR vs R test ($p=0.82$) confirmed that adding dexmedetomidine does not appreciably increase hemorrhage risk.

8 Full Time Courses

8.1 Single Horse Timeline

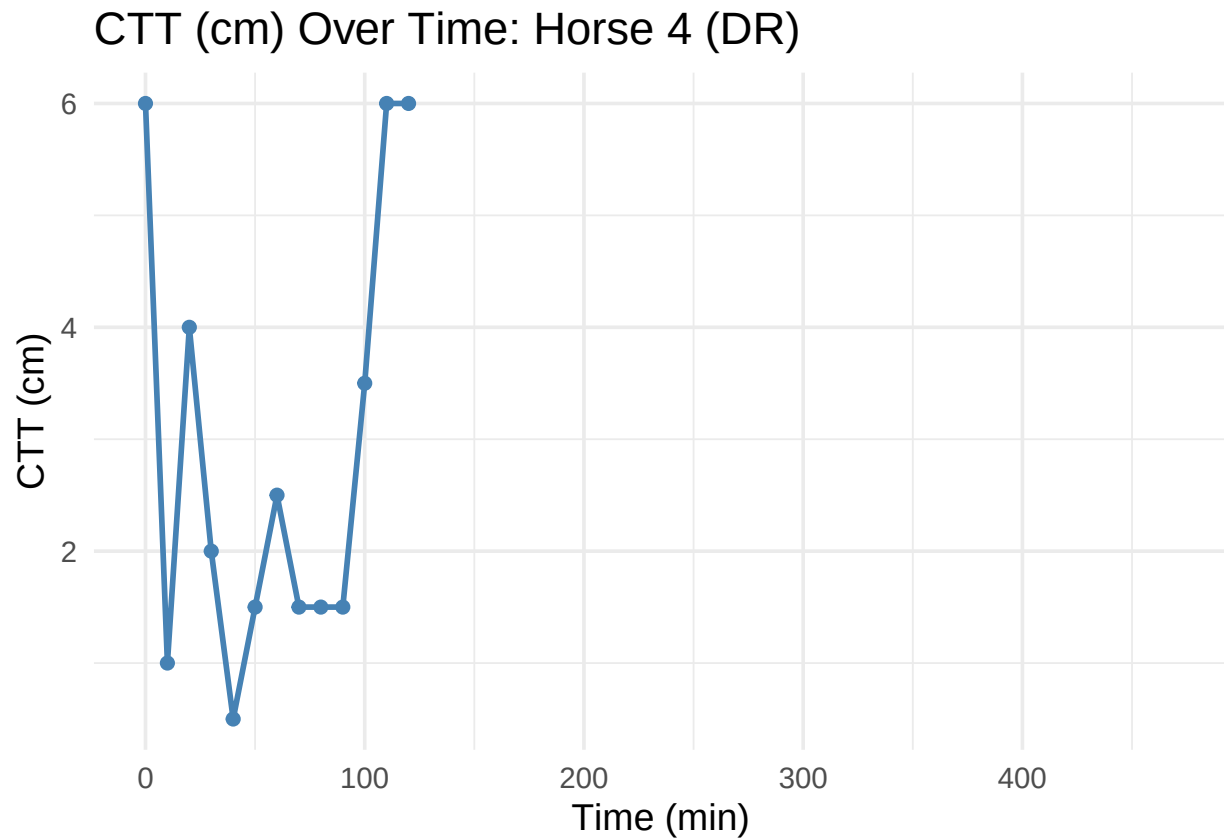
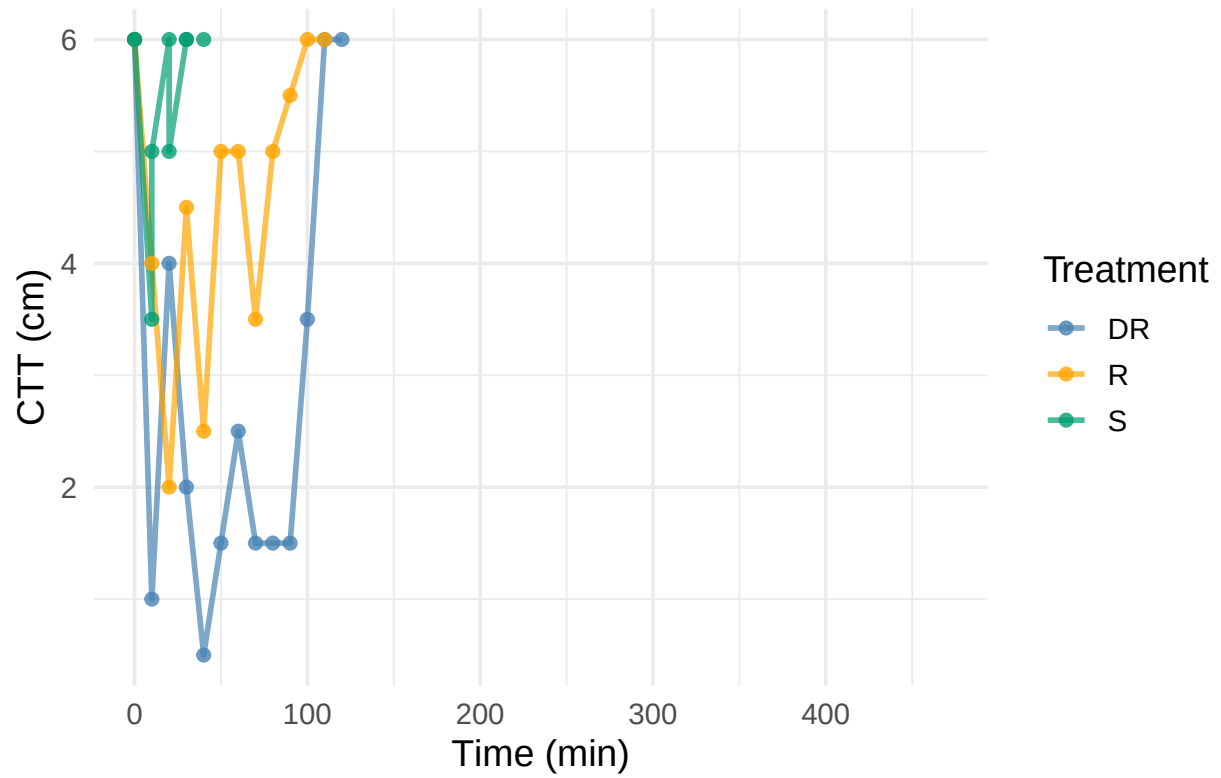


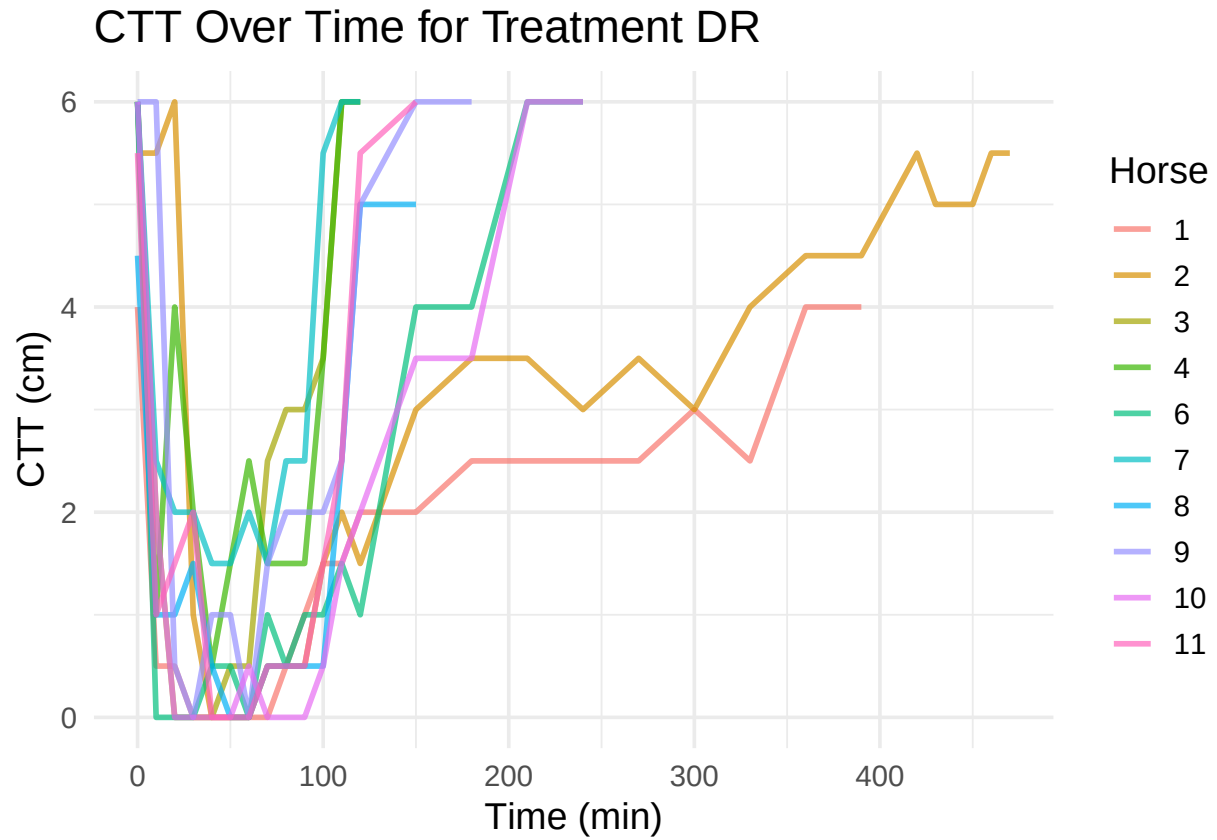
Table 19: Summarized Metrics for Horse 4 (Treatment DR)

Metric	Value
Minimum CTT (cm)	0.5
Duration of Min CTT (min)	10.0
Total Anesthesia Time (min)	110.0

CTT Over Time: Horse 4 Across Treatments

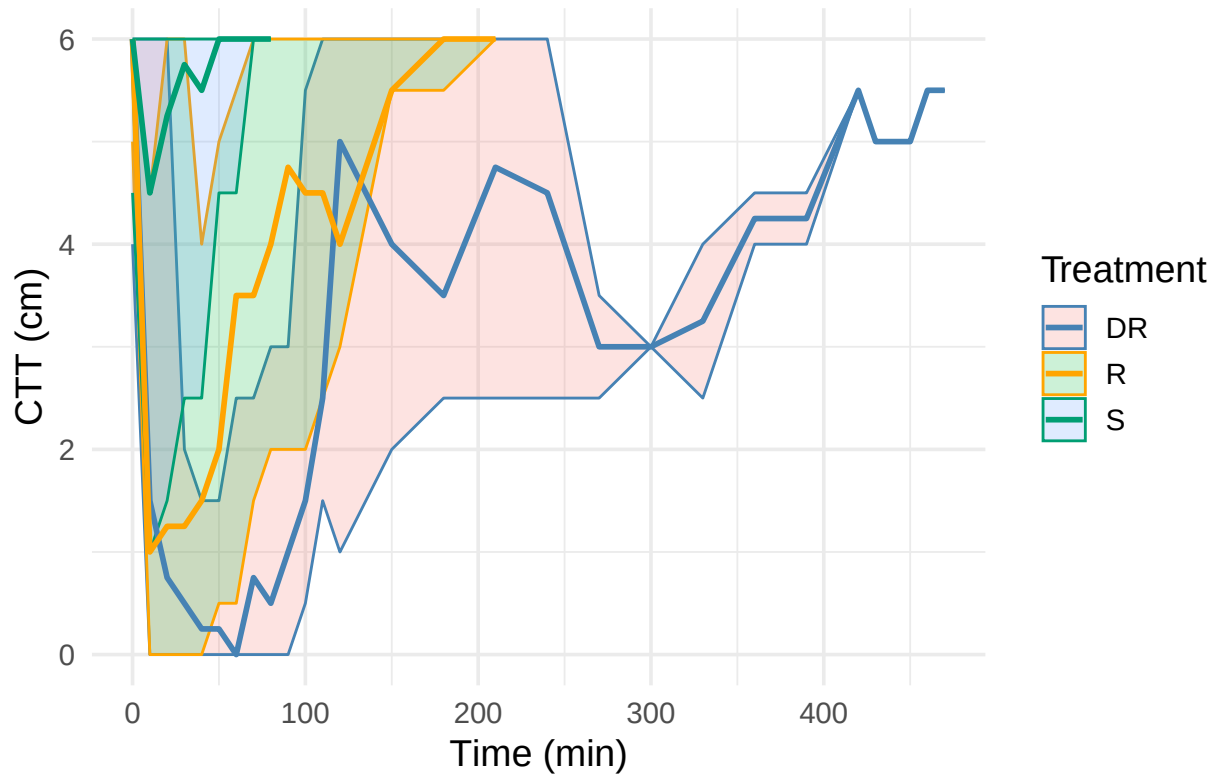


8.2 All Horse Timeline

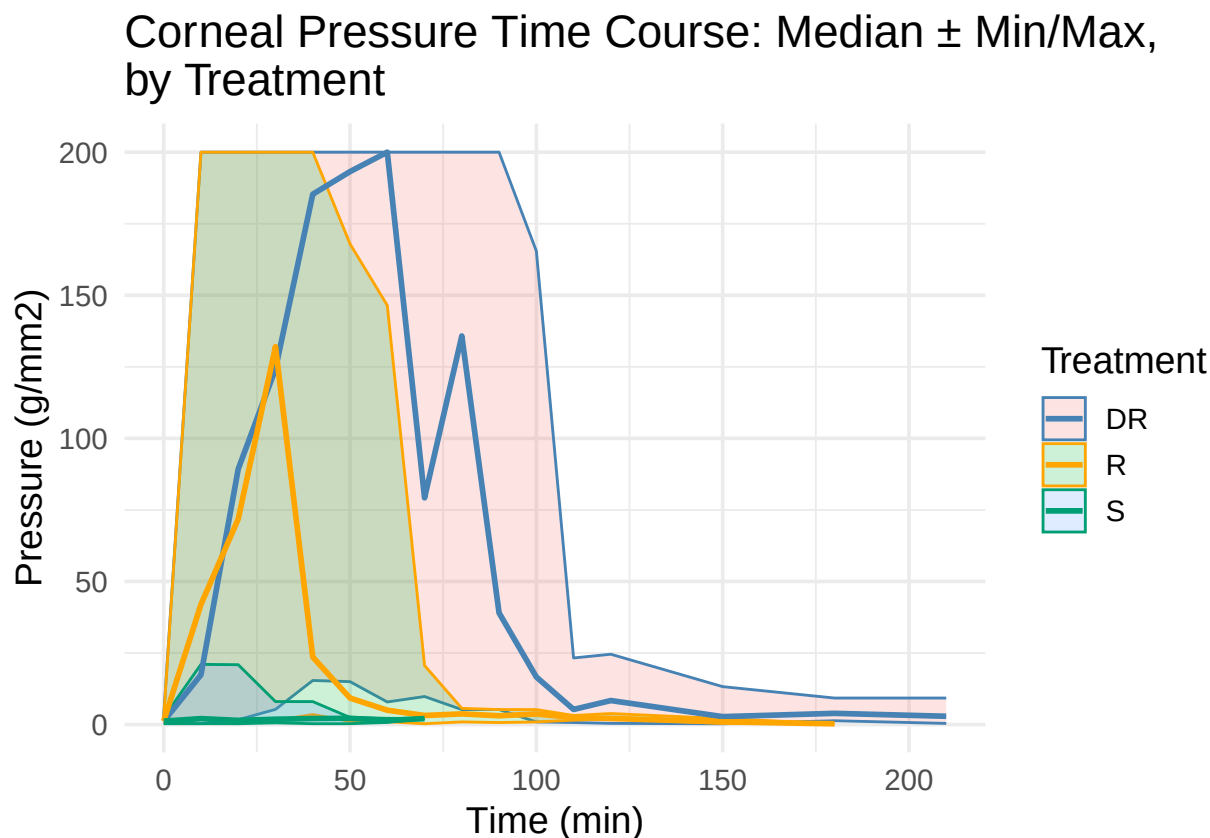


8.3 CCT and Corneal Pressure time course, ALL TREATMENTS

CTT Time Course: Median $\pm$ Min/Max by Treatment



8.4 Corneal Pressure Time Course for One Horse, All Treatments



Section 8 begins with a detailed “single-horse timeline” of CTT for Horse 8 under DR (plot 8.1 and its accompanying summary table), showing the rapid drop to zero sensitivity, a sustained plateau at maximal anesthesia, and the eventual return to baseline. Next, the CTT curves for that same horse across all three treatments make it clear that DR produces both the deepest blockade and the longest recovery compared to R and saline. Additionally, plot 8.2 overlays every horse’s CTT curve under the DR treatment and clearly shows that, under dexmedetomidine + ropivacaine, most animals maintain near-zero corneal touch thresholds for well over two hours. A few recover slightly earlier and out two outliers way later – highlighting both the uniformly prolonged blockade and the remaining individual variability. We then move to group-level summaries: plot 8.3 presents the median $\pm$ min/max CTT over time for each treatment, illustrating that DR maintains a median filamental length of zero longer and with tighter bounds than R or S, while saline rebounds almost immediately. Finally, plot 8.4 offers the same median $\pm$ min/max ribbon format for corneal pressure, confirming that DR also sustains reduced tonometry readings farther and more consistently than the other treatments. Together, these four graphics—from individual trajectories to aggregated ribbons—visually reinforce that dexmedetomidine + ropivacaine outperforms ropivacaine alone and saline in both depth and uniformity of corneal anesthesia.

9 Appendix

```
knitr::opts_chunk$set(  
  echo = FALSE,  
  warning = FALSE,
```

```

  message = FALSE
)
# Load libraries
library(knitr)
library(tidyverse)
library(openxlsx)
library(ggplot2)
library(janitor)
library(dplyr)
library(scales)
library(lme4)
library(lmerTest)
library(emmeans)
library(kableExtra)
library(broom.mixed)
library(broom)
#making original data frame (without horse_period)
df_summ <- read.xlsx("Ropivacaine_Dexmed_Data_AH.xlsx", sheet = "Summarized") %>%
  rename(
    horse           = Horse,
    treatment       = Treatment,
    period          = Period,
    minimum_ctt_cm  = `Minimum.CTT.(cm)`,
    duration_of_min_ctt_min = `Duration.of.Min.CTT.(min)`,
    maximum_corneal_pressure = Maximum.corneal.pressure,
    total_anesthesia_time_min = `Total.Anesthesia.Time.(min)`
  ) %>%
  mutate(
    horse      = factor(horse),
    treatment = factor(treatment, levels = c("DR", "R", "S")),
    period     = factor(period)
  )

#adding horse_period
df_summ2 <- df_summ %>%
  mutate(
    horse_period = interaction(horse, period, sep = "_")
  )

#adding HemScore
#load the hemorrhage-score sheet, keeping only horse, treatment, and score
df_hem_raw <- read.xlsx("Ropivacaine_Dexmed_Data_AH.xlsx", sheet = "HemScore") %>%
  clean_names()
#align types and fill missing
df_hem <- df_hem_raw %>%
  mutate(
    horse      = factor(horse, levels = levels(df_summ2$horse)),
    treatment = factor(treatment, levels = levels(df_summ2$treatment)),
    hem_score = replace_na(hemorrhage_score, 0)
  ) %>%
  select(horse, treatment, hem_score)

#join hem_score onto summarized data

```

```

df_summ3 <- df_summ2 %>%
  left_join(df_hem, by = c("horse", "treatment")) %>%
  mutate(
    hem_score = replace_na(hem_score, 0)
  )

#loading in the full dataset
#read in full sheet from xlsx file
df_full <- read.xlsx("Ropivacaine_Dexmed_Data_AH.xlsx", sheet = "Full") %>%
  #drop any completely empty rows
  filter(!is.na(Variable))

#choosing horse (THIS ALLOWS FOR VERY EASY MANIPULATION AND VIEWING FOR HORSES AND TREATMENTS)
sel_horse <- "4"
sel_trt <- "DR"

#spaghetti plot
spaghetti_horse <- ggplot(df_summ3, aes(x = treatment, y = total_anesthesia_time_min, group = horse, color = horse)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  labs(
    title = "Total Anesthesia Time by Treatment",
    subtitle = "Lines grouped by horse",
    x = "Treatment",
    y = "Total Anesthesia Time (min)",
    color = "Horse ID"
  ) +
  theme_minimal() +
  theme(legend.position = "right")
print(spaghetti_horse)

#hemscore distribution plot
hem_score_barplot <- ggplot(df_summ3, aes(x = treatment, fill = factor(hem_score))) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_brewer("Hem Score", palette = "Reds") +
  labs(
    x = "Treatment",
    y = "Proportion of Eyes",
    title = "Distribution of Hemorrhage Scores by Treatment"
  ) +
  theme_minimal()

print(hem_score_barplot)
#####
#
# RESPONSE VARIABLE 1: TOTAL ANESTHESIA TIME #
#
#####

#creating summary table
df_summ2 %>%

```

```

group_by(treatment) %>%
  summarise(
    N      = n(),
    Mean   = mean(total_anesthesia_time_min),
    SD     = sd(total_anesthesia_time_min),
    Min    = min(total_anesthesia_time_min),
    Median = median(total_anesthesia_time_min),
    Max    = max(total_anesthesia_time_min)
  ) %>%
  kable(
    caption = "Summary: Total Anesthesia Time by Treatment",
    digits  = 1
  ) %>%
  kable_styling(full_width = FALSE)
#fitting lm model for sqrt transformed tta
m_tta_hp <- lmer(
  sqrt(total_anesthesia_time_min) ~ treatment +
    (1 | horse),
  data  = df_summ2,
  REML  = TRUE
)

# Type III ANOVA
anova_tbl_sqrt_hp <- as.data.frame(anova(m_tta_hp))
anova_tbl_sqrt_hp %>%
  rownames_to_column("Term") %>%
  select(
    Term,
    `Num DF`  = NumDF,
    `Den DF`  = DenDF,
    `F value` = `F value`,
    `Pr(>F)`  = `Pr(>F)`
  ) %>%
  mutate(
    `Pr(>F)` = if_else(
      `Pr(>F)` < 1e-4,
      formatC(`Pr(>F)`, format = "e", digits = 2), # scientific for tiny p-values
      sprintf("%.4f", `Pr(>F)`) # 4 decimal places otherwise
    )
  ) %>%
  kable(
    caption = "Type III F-test for sqrt(Total Anesthesia Time)",
    digits  = c(NA, 0, 0, 3, 6) # + last entry = 6 decimals for Pr(>F)
  ) %>%
  kable_styling(full_width = FALSE)
#running emmeans
emm_sqrt_tta_hp <- emmeans(m_tta_hp, pairwise ~ treatment, adjust = "tukey")

emm_sqrt_tta_hp$contrasts %>%
  as.data.frame() %>%
  mutate(
    p_value_fmt = if_else(

```

```

    p.value < 1e-4,
    formatC(p.value, format = "e", digits = 2),
    sprintf("%.5f", p.value)
  )
) %>%
select(Contrast = contrast,
       Estimate = estimate,
       SE,
       DF = df,
       "T-Ratio" = t.ratio,
       "P-Value" = p_value_fmt) %>%
kable(
  digits = c(NA, 3, 2, 0, 2, 4),
  format = "latex",
  caption = "Tukey Pairwise Contrasts for Treatment (Sqrt Transformed Total Anesthesia Time (Units: $))",
  longtable = FALSE
) %>%
kable_styling(full_width = TRUE,
               latex_options = c("hold_position", "scale_down"))

# emmip plot
emmip(emm_sqrt_tta_hp, ~ treatment, CI = TRUE, type = "response") +
aes(color = treatment, fill = treatment) +
  scale_color_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  ) +
  scale_fill_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  )
)

#residual v fitted plot
resid_sqrt_tta_hp <- tibble(
  fitted = fitted(m_tta_hp),
  resid = resid(m_tta_hp)
)

ggplot(resid_sqrt_tta_hp, aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  labs(
    x = "Fitted values",
    y = "Residuals",
    title = "Residuals vs. Fitted for Response = Sqrt Transformed Total Anesthesia Time",
    subtitle = expression(paste("Response Units: min"))
  ) +
  theme_minimal(base_size = 12)

```



```

#QQ plot
ggplot(data = tibble(resid = resid(m_tta_hp)), aes(sample = resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "QQ-plot of Residuals for Response = Sqrt Transformed Total Anesthesia Time",
    subtitle = expression(paste("Response Units: min")),
    x      = "Theoretical Quantiles",
    y      = "Sample Quantiles"
  ) +
  theme_minimal(base_size = 12)
#paired DR vs R on df_summ2, ensuring complete pairs
cts_tta <- df_summ2 %>%
  filter(treatment %in% c("DR", "R")) %>%
  select(horse, treatment, total_anesthesia_time_min) %>%
  pivot_wider(
    names_from = treatment,
    values_from = total_anesthesia_time_min
  ) %>%
  filter(!is.na(DR) & !is.na(R))

#run Wilcoxon signed-rank test with Hodges-Lehmann CI
wil_tta <- wilcox.test(
  cts_tta$DR,
  cts_tta$R,
  paired    = TRUE,
  exact     = FALSE,
  conf.int  = TRUE,
  conf.level = 0.95
)

#summarize test results
tibble(
  Statistic      = wil_tta$statistic,
  `HL Estimate`  = wil_tta$estimate,
  `95% Lower`    = wil_tta$conf.int[1],
  `95% Upper`    = wil_tta$conf.int[2],
  `p-value`      = wil_tta$p.value
) %>%
  kable(
    caption = "Wilcoxon Test: Total Anesthesia Time (DR vs R)",
    digits  = c(0,0,0,0,4)
  ) %>%
  kable_styling(full_width = FALSE)
#####
#                                     #
# Response Variable 2 = Minimum CTT (cm) #
#                                     #
#####

#summary table

sum_mcct <- df_summ2 %>%

```

```

group_by(Treatment = treatment) %>%
  summarize(
    N = n(),
    Mean = mean(minimum_ctt_cm, na.rm = TRUE),
    SD = sd(minimum_ctt_cm, na.rm = TRUE),
    Min = min(minimum_ctt_cm, na.rm = TRUE),
    Median = median(minimum_ctt_cm, na.rm = TRUE),
    Max = max(minimum_ctt_cm, na.rm = TRUE),
    .groups = "drop"
  )

sum_mcct %>%
  kable(
    digits = 2,
    format = "latex",
    caption = "Summary of Minimum CTT (Units: $cm$) by Treatment",
    longtable = FALSE
  ) %>%
  kable_styling(full_width = FALSE,
                latex_options = c("hold_position", "scale_down"))
#fitting model

m_mcct <- lmer(
  minimum_ctt_cm ~ treatment +
    (1 | horse),
  data = df_summ2
)

# capture the ANOVA table as a data.frame
anova_tbl_mcct <- as.data.frame(anova(m_mcct))

anova_tbl_mcct %>%
  rownames_to_column("Term") %>%
  select(
    Term,
    `Num DF` = NumDF,
    `Den DF` = DenDF,
    `F value` = `F value`,
    `Pr(>F)` = `Pr(>F)`
  ) %>%
  mutate(
    `Pr(>F)` = if_else(
      `Pr(>F)` < 1e-4,
      formatC(`Pr(>F)`, format = "e", digits = 2), # scientific for tiny p-values
      sprintf("%.4f", `Pr(>F)`), # 4 decimal places otherwise
    )
  ) %>%
  kable(
    caption = "Type III F-test for Minimum CCT",
    digits = c(NA, 0, 0, 3, 6) # + last entry = 6 decimals for Pr(>F)
  ) %>%
  kable_styling(full_width = FALSE)

```

```

#running emmeans
emm_mcct <- emmeans(m_mcct, pairwise ~ treatment, adjust = "tukey")

emm_mcct$contrasts %>%
  as.data.frame() %>%
  mutate(
    p_value_fmt = if_else(
      p.value < 1e-4,
      formatC(p.value, format = "e", digits = 2),
      sprintf("%.5f", p.value)
    )
  ) %>%
  select(Contrast = contrast,
         Estimate = estimate,
         SE,
         DF = df,
         "T-Ratio" = t.ratio,
         "P-Value" = p_value_fmt) %>%
  kable(
    digits = c(NA, 3, 2, 0, 2, 4),
    format = "latex",
    caption = "Tukey Pairwise Contrasts for Treatment (Minimum CCT (Units: $cm$))",
    longtable = FALSE
  ) %>%
  kable_styling(full_width = TRUE,
                latex_options = c("hold_position", "scale_down"))

# emmip plot
emmip(emm_mcct, ~ treatment, CI = TRUE, type = "response") +
  aes(color = treatment, fill = treatment) +
  scale_color_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  ) +
  scale_fill_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  )
)

#residual v fitted plot
resid_mcct <- tibble(
  fitted = fitted(m_mcct),
  resid = resid(m_mcct)
)

ggplot(resid_mcct, aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +

```

```

labs(
  x = "Fitted values",
  y = "Residuals",
  title = "Residuals vs. Fitted for Response = Minimum CTT",
  subtitle = expression(paste("Response Units: cm"))
) +
theme_minimal(base_size = 12)

#QQ plot
ggplot(data = tibble(resid = resid(m_mcct)), aes(sample = resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "QQ-plot of Residuals for Response = Minimum CTT",
    subtitle = expression(paste("Response Units: cm")),
    x = "Theoretical Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_minimal(base_size = 12)

# pivot to wide for DR and R
wide_minctt <- df_summ %>%
  filter(treatment %in% c("DR", "R")) %>%
  select(horse, treatment, minimum_ctt_cm) %>%
  pivot_wider(names_from = treatment, values_from = minimum_ctt_cm)

# Run paired Wilcoxon test
wil_mcct <- wilcox.test(wide_minctt$DR, wide_minctt$R,
  paired = TRUE, exact = FALSE,
  conf.int = TRUE,
  conf.level = 0.95)

#summarize results
tibble(
  Statistic = wil_mcct$statistic,
  `H-L Estimate` = wil_mcct$estimate,
  `95% Lower` = wil_mcct$conf.int[1],
  `95% Upper` = wil_mcct$conf.int[2],
  `p-value` = wil_mcct$p.value
) %>%
  kable(
    caption = "Wilcoxon Signed-Rank Test with Hodges-Lehmann CI",
    digits = c(0, 0, 0, 0, 4)
  ) %>%
  kable_styling(full_width = FALSE)
#####
# #
# RESPONSE VARIABLE 3: DURATION OF MINIMUM CTT (mins) #
# #
#####

#summary table

```

```

sum_dmcct <- df_summ2 %>%
  group_by(Treatment = treatment) %>%
  summarize(
    N = n(),
    Mean = mean(duration_of_min_ctt_min, na.rm = TRUE),
    SD = sd(duration_of_min_ctt_min, na.rm = TRUE),
    Min = min(duration_of_min_ctt_min, na.rm = TRUE),
    Median = median(duration_of_min_ctt_min, na.rm = TRUE),
    Max = max(duration_of_min_ctt_min, na.rm = TRUE),
    .groups = "drop"
  )

sum_dmcct %>%
  kable(
    digits = 2,
    format = "latex",
    caption = "Summary of Duration of Minimum CTT (Units: $min$) by Treatment",
    longtable = FALSE
  ) %>%
  kable_styling(full_width = FALSE,
                latex_options = c("hold_position", "scale_down"))
#fitting model

m_dmcct <- lmer(
  duration_of_min_ctt_min ~ treatment +
    (1 | horse),
  data = df_summ2
)

# capture the ANOVA table as a data.frame
anova_tbl_dmcct <- as.data.frame(anova(m_dmcct))

anova_tbl_dmcct %>%
  rownames_to_column("Term") %>%
  select(
    Term,
    `Num DF` = NumDF,
    `Den DF` = DenDF,
    `F value` = `F value`,
    `Pr(>F)` = `Pr(>F)`
  ) %>%
  mutate(
    `Pr(>F)` = if_else(
      `Pr(>F)` < 1e-4,
      formatC(`Pr(>F)`, format = "e", digits = 2), # scientific for tiny p-values
      sprintf("%.4f", `Pr(>F)`), # 4 decimal places otherwise
    )
  ) %>%
  kable(
    caption = "Type III F-test for Duration of Minimum CCT",
    digits = c(NA, 0, 0, 3, 6) # + last entry = 6 decimals for Pr(>F)
  ) %>%
  kable_styling(full_width = FALSE)

```

```

#running emmeans
emm_dmcct <- emmeans(m_dmcct, pairwise ~ treatment, adjust = "tukey")

emm_dmcct$contrasts %>%
  as.data.frame() %>%
  mutate(
    p_value_fmt = if_else(
      p.value < 1e-4,
      formatC(p.value, format = "e", digits = 2),
      sprintf("%.5f", p.value)
    )
  ) %>%
  select(Contrast = contrast,
         Estimate = estimate,
         SE,
         DF = df,
         "T-Ratio" = t.ratio,
         "P-Value" = p_value_fmt) %>%
  kable(
    digits = c(NA, 3, 2, 0, 2, 4),
    format = "latex",
    caption = "Tukey Pairwise Contrasts for Treatment (Duration Minimum CCT (Units: $min$))",
    longtable = FALSE
  ) %>%
  kable_styling(full_width = TRUE,
                latex_options = c("hold_position", "scale_down"))

# emmip plot
emmip(emm_dmcct, ~ treatment, CI = TRUE, type = "response") +
  aes(color = treatment, fill = treatment) +
  scale_color_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  ) +
  scale_fill_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  )
)

#residual v fitted plot
resid_dmcct <- tibble(
  fitted = fitted(m_dmcct),
  resid = resid(m_dmcct)
)

ggplot(resid_dmcct, aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +

```

```

labs(
  x = "Fitted values",
  y = "Residuals",
  title = "Residuals vs. Fitted for Response = Duration Minimum CCT (min)",
  subtitle = expression(paste("Response Units: mins"))
) +
theme_minimal(base_size = 12)

#QQ plot
ggplot(data = tibble(resid = resid(m_dmcct)), aes(sample = resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "QQ-plot of Residuals for Response = Duration Minimum CCT (min)",
    subtitle = expression(paste("Response Units: mins")),
    x = "Theoretical Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_minimal(base_size = 12)

# pivot to wide for DR and R
wide_dmctt <- df_summ %>%
  filter(treatment %in% c("DR", "R")) %>%
  select(horse, treatment, duration_of_min_ctt_min) %>%
  pivot_wider(names_from = treatment, values_from = duration_of_min_ctt_min)

# Run paired Wilcoxon test
wil_dmcct <- wilcox.test(wide_dmctt$DR, wide_dmctt$R,
  paired = TRUE, exact = FALSE,
  conf.int = TRUE,
  conf.level = 0.95)

#summarize results
tibble(
  Statistic = wil_dmcct$statistic,
  `H-L Estimate` = wil_dmcct$estimate,
  `95% Lower` = wil_dmcct$conf.int[1],
  `95% Upper` = wil_dmcct$conf.int[2],
  `p-value` = wil_dmcct$p.value
) %>%
  kable(
    caption = "Wilcoxon Signed-Rank Test with Hodges-Lehmann CI",
    digits = c(0, 0, 0, 0, 4)
  ) %>%
  kable_styling(full_width = FALSE)
#####
# #
# RESPONSE VARIABLE 4: MAX CORNEAL PRESSURE #
# #
#####

#summary table

```

```

sum_mp <- df_summ2 %>%
  group_by(Treatment = treatment) %>%
  summarize(
    N = n(),
    Mean = mean(maximum_corneal_pressure, na.rm = TRUE),
    SD = sd(maximum_corneal_pressure, na.rm = TRUE),
    Min = min(maximum_corneal_pressure, na.rm = TRUE),
    Median = median(maximum_corneal_pressure, na.rm = TRUE),
    Max = max(maximum_corneal_pressure, na.rm = TRUE),
    .groups = "drop"
  )

sum_mp %>%
  kable(
    digits = 2,
    format = "latex",
    caption = paste("Summary of Maximum Corneal Pressure ( $\text{g/mm}^2$ ) by Treatment"),
    longtable = FALSE,
    escape = FALSE
  ) %>%
  kable_styling(full_width = FALSE,
                latex_options = c("hold_position", "scale_down"))
#fitting model

m_mp <- lmer(
  maximum_corneal_pressure ~ treatment +
    (1 | horse),
  data = df_summ2
)

# capture the ANOVA table as a data.frame
anova_tbl_mp <- as.data.frame(anova(m_mp))

anova_tbl_mp %>%
  rownames_to_column("Term") %>%
  select(
    Term,
    `Num DF` = NumDF,
    `Den DF` = DenDF,
    `F value` = `F value`,
    `Pr(>F)` = `Pr(>F)`
  ) %>%
  mutate(
    `Pr(>F)` = if_else(
      `Pr(>F)` < 1e-4,
      formatC(`Pr(>F)`, format = "e", digits = 2), # scientific for tiny p-values
      sprintf("%.4f", `Pr(>F)`), # 4 decimal places otherwise
    )
  ) %>%
  kable(
    caption = "Type III F-test for Max Corneal Pressure",
    digits = c(NA, 0, 0, 3, 6) # + last entry = 6 decimals for Pr(>F)
  ) %>%

```



```

kable_styling(full_width = FALSE)
#running emmeans
emm_mp <- emmeans(m_mp, pairwise ~ treatment, adjust = "tukey")

emm_mp$contrasts %>%
  as.data.frame() %>%
  mutate(
    p_value_fmt = if_else(
      p.value < 1e-4,
      formatC(p.value, format = "e", digits = 2),
      sprintf("%.5f", p.value)
    )
  ) %>%
  select(Contrast = contrast,
         Estimate = estimate,
         SE,
         DF = df,
         "T-Ratio" = t.ratio,
         "P-Value" = p_value_fmt) %>%
  kable(
    digits = c(NA, 3, 2, 0, 2, 4),
    format = "latex",
    caption = "Tukey Pairwise Contrasts for Treatment (Maximum Corneal Pressure (Units: $g/mm^2$))",
    longtable = FALSE,
    escape = FALSE
  ) %>%
  kable_styling(full_width = TRUE,
                latex_options = c("hold_position", "scale_down"))

# emmip plot
emmip(emm_mp, ~ treatment, CI = TRUE, type = "response") +
aes(color = treatment, fill = treatment) +
  scale_color_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  ) +
  scale_fill_manual(
    values = c(
      DR = "steelblue",
      R = "orange",
      S = "#009E73"
    )
  )
)

#residual v fitted plot
resid_mp <- tibble(
  fitted = fitted(m_mp),
  resid = resid(m_mp)
)

ggplot(resid_mp, aes(x = fitted, y = resid)) +

```

```

geom_point(alpha = 0.5) +
geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
labs(
  x = "Fitted values",
  y = "Residuals",
  title = "Residuals vs. Fitted for Response = Maximum Corneal Pressure",
  subtitle = expression(paste("Response Units: ", g/mm^2))
) +
theme_minimal(base_size = 12)

#QQ plot
ggplot(data = tibble(resid = resid(m_mp)), aes(sample = resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "QQ-plot of Residuals for Response = Maximum Corneal Pressure",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles",
    subtitle = expression(paste("Response Units: ", g/mm^2))
  ) +
  theme_minimal(base_size = 12)

# pivot to wide for DR and R
wide_mp <- df_summ %>%
  filter(treatment %in% c("DR", "R")) %>%
  select(horse, treatment, maximum_corneal_pressure) %>%
  pivot_wider(names_from = treatment, values_from = maximum_corneal_pressure)

# Run paired Wilcoxon test
wil_mp <- wilcox.test(wide_mp$DR, wide_mp$R,
  paired = TRUE, exact = FALSE,
  conf.int = TRUE,
  conf.level = 0.95)

#summarize results
tibble(
  Statistic = wil_mp$statistic,
  `H-L Estimate` = wil_mp$estimate,
  `95% Lower` = wil_mp$conf.int[1],
  `95% Upper` = wil_mp$conf.int[2],
  `p-value` = wil_mp$p.value
) %>%
  kable(
    caption = "Wilcoxon Signed-Rank Test with Hodges-Lehmann CI",
    digits = c(0, 0, 0, 0, 4)
  ) %>%
  kable_styling(full_width = FALSE)
#####
#
# RESPONSE VARIABLE : HEMORRHAGE SCORE #
#
#####

```

```

df_summ3 %>%
  group_by(treatment) %>%
  summarise(
    N = n(),
    Median = median(hem_score, na.rm = TRUE),
    Mean = mean(hem_score, na.rm = TRUE),
    SD = sd(hem_score, na.rm = TRUE),
    Min = min(hem_score, na.rm = TRUE),
    Max = max(hem_score, na.rm = TRUE)
  ) %>%
  kable(
    caption = "Summary: Hemorrhage Score by Treatment",
    digits = c(0,1,1,2,0,0)
  ) %>%
  kable_styling(full_width = FALSE)
#pivot to wide for DR and R
wide_hem <- df_summ3 %>%
  filter(treatment %in% c("DR", "R")) %>%
  select(horse, treatment, hem_score) %>%
  pivot_wider(names_from = treatment, values_from = hem_score)

#run paired Wilcoxon test
wil_hem <- wilcox.test(wide_hem$DR, wide_hem$R,
  paired = TRUE, exact = FALSE,
  conf.int = TRUE,
  conf.level = 0.95)

#summarize results
tibble(
  Statistic = wil_hem$statistic,
  `H-L Estimate` = wil_hem$estimate,
  `95% Lower` = wil_hem$conf.int[1],
  `95% Upper` = wil_hem$conf.int[2],
  `p-value` = wil_hem$p.value
) %>%
  kable(
    caption = "Wilcoxon Signed-Rank Test with Hodges-Lehmann CI",
    digits = c(0, 0, 0, 0, 4)
  ) %>%
  kable_styling(full_width = FALSE)
#####
#
# TIME COURSE FOR ONE HORSE, ONE TREATMENT #
#
#####

#making subset for one horse (and changing CTT to long form)
df_one_ctt <- df_full %>%
  filter(Horse == sel_horse, Treatment == sel_trt, Variable == "CTT (cm)") %>%
  pivot_longer(
    cols = c('Time.-.Baseline', matches("[0-9]+$")),

```

```

    names_to = "time_label",
    values_to = "ctt_cm"
  ) %>%
  mutate(
    #parse whatever digits are in the label
    time_min = parse_number(time_label),
    #replace that NA with 0
    time_min = replace_na(time_min, 0)
  )

#plotting
ggplot(df_one_ctt, aes(x = time_min, y = ctt_cm)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "steelblue", size = 2) +
  labs(
    title = paste0("CTT (cm) Over Time: Horse ", sel_horse, " (", sel_trt, ")"),
    x      = "Time (min)",
    y      = "CTT (cm)"
  ) +
  theme_minimal(base_size = 14)

#associated table
#pull the matching summarized row into a new object
example_summary_matched2 <- df_summ2 %>%
  filter(horse == sel_horse, treatment == sel_trt) %>%
  select(
    minimum_ctt_cm,
    duration_of_min_ctt_min,
    total_anesthesia_time_min
  )

#pivot longer so we have Metric / Value
example_summary_matched2_long <- example_summary_matched2 %>%
  pivot_longer(
    cols      = everything(),
    names_to  = "Metric",
    values_to = "Value"
  ) %>%
  mutate(
    Metric = recode(Metric,
      minimum_ctt_cm      = "Minimum CTT (cm)",
      duration_of_min_ctt_min = "Duration of Min CTT (min)",
      total_anesthesia_time_min = "Total Anesthesia Time (min)"
    )
  )

#print it
example_summary_matched2_long %>%
  kable(
    col.names = c("Metric", "Value"),
    caption   = paste0(
      "Summarized Metrics for Horse ",
      sel_horse, " (Treatment ", sel_trt, ")"
    )
  )

```

```

)
) %>%
kable_styling(
  bootstrap_options = c("striped", "hover"),
  full_width        = FALSE,
  font_size          = 12,
  latex_options     = c("hold_position", "scale_down")
)
#####
#                                     #
# TIME COURSE FOR ONE HORSE, ALL TREATMENTS #
#                                     #
#####

#creating long df of full data
df_full_long <- df_full %>%
  pivot_longer(
    cols      = -c(Horse, Treatment, Variable),
    names_to   = "time_label",
    values_to  = "value"
  ) %>%
  mutate(
    time_min = replace_na(suppressWarnings(as.numeric(time_label))), 0)
  ) %>%
  filter(!is.na(time_min))

# grab CTT for this one horse, all treatments
df_one_horse_ctt <- df_full_long %>%
  filter(
    Horse      == sel_horse,
    Variable    == "CTT (cm)"
  )

ggplot(df_one_horse_ctt, aes(x = time_min, y = value, color = Treatment)) +
  geom_line(size = 1, alpha = 0.7) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_manual(
    values = c(
      DR = "steelblue",
      R  = "orange",
      S  = "#009E73"
    )
  ) +
  labs(
    title = paste0("CTT Over Time: Horse ", sel_horse, " Across Treatments"),
    x      = "Time (min)",
    y      = "CTT (cm)",
    color  = "Treatment"
  ) +
  theme_minimal(base_size = 14)
# Overlay CTT timelines for all horses for the selected treatment
df_full_long %>%
  filter(Variable == "CTT (cm)", Treatment == sel_trt) %>%

```

```

ggplot(aes(
  x      = time_min,
  y      = value,
  group  = Horse,
  color  = as.factor(Horse)
)) +
geom_line(alpha = 0.7, size = 1) +
labs(
  title = paste0("CTT Over Time for Treatment ", sel_trt),
  x      = "Time (min)",
  y      = "CTT (cm)",
  color  = "Horse"
) +
theme_minimal(base_size = 14)
#####
#
# CTT AND CORNEAL PRESSURE TIME COURSE FOR 1 HORSE, ALL TREATMENTS #
#
#####

#summarize just the CTT data
df_ctt_timecourse <- df_full_long %>%
  filter(Variable == "CTT (cm)", !is.na(value)) %>%
  group_by(time_min, Treatment) %>%
  summarise(
    med = median(value, na.rm = TRUE),
    lo  = min(value,    na.rm = TRUE),
    hi  = max(value,    na.rm = TRUE),
    .groups = "drop"
  )

#plot
ggplot(df_ctt_timecourse, aes(x = time_min, y = med, color = Treatment, fill = Treatment)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2) +
  geom_line(size = 1) +
  scale_color_manual(
    values = c(
      DR = "steelblue",
      R  = "orange",
      S  = "#009E73"
    )
  ) +
  labs(
    title = "CTT Time Course: Median ± Min/Max by Treatment",
    x      = "Time (min)",
    y      = "CTT (cm)"
  ) +
  theme_minimal(base_size = 14)

#summarize just the Corneal Pressure data
df_cp_timecourse <- df_full_long %>%
  filter(Variable == "Corneal Pressure (g/mm2)", !is.na(value)) %>%
  group_by(time_min, Treatment) %>%

```

```

summarise(
  med = median(value, na.rm = TRUE),
  lo  = min(value,    na.rm = TRUE),
  hi  = max(value,    na.rm = TRUE),
  .groups = "drop"
)

#plot
ggplot(df_cp_timecourse, aes(x = time_min, y = med, color = Treatment, fill = Treatment)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2) +
  geom_line(size = 1) +
  scale_color_manual(
    values = c(
      DR = "steelblue",
      R  = "orange",
      S  = "#009E73"
    )
  ) +
  labs(
    title = "Corneal Pressure Time Course: Median ± Min/Max, \nby Treatment",
    x      = "Time (min)",
    y      = "Pressure (g/mm2)"
  ) +
  theme_minimal(base_size = 14)

```